26.

Geometry

Type - 1

1. If the angles of the triangle are in the ratio 1:2:3, then what is the smallest angle?

(a) 40° (b) 25°
(c) 15° (d) 30°

RRB RPF Constable – 17/01/2019 (Shift-I)

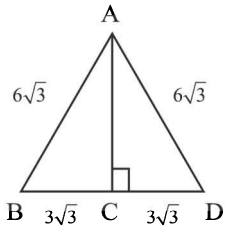
Ans : (d) Let angles of triangle are x, 2x, 3x then–
sum of three angles of triangle = 180°
x + 2x + 3x = 180°
6 x = 180°
$\boxed{x = 30°}$
So smallest angle will be 30°

2. If one side of an equilateral triangle is $6\sqrt{3}$ cm, then what is its hight?

(a) 9 cm (b) 6 cm
(c) 3 cm (d) $3\sqrt{3}$ cm

RRB RPF SI – 13/01/2019 (Shift-II)

Ans : (a) Side of equilateral triangle = $6\sqrt{3}$



in triangle ABC -

$(AC)^2 = (AB)^2 - (BC)^2$

$= \left(6\sqrt{3}\right)^2 - \left(3\sqrt{3}\right)^2 = 108 - 27$

$(AC)^2 = 81$

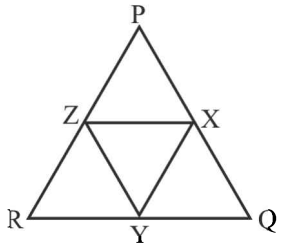
height = 9 cm

3. The midpoints of the sides of an equilateral triangle PQR are X, Y and Z. If the circumference of triangle PQR is 24 cm, then what will be the circumference of traingle XYZ?

(a) 96 cm (b) 36 cm
(c) 12 cm (d) 48 cm

RRB RPF Constable – 22/01/2019 (Shift-II)

Ans : (c) According to the question-
PQ + QR + PR = 24 cm



In a triangle, the line joining the midpoints of two sides is parallel and half of the third side.

$ZX = \frac{1}{2}$ QR ..... (i)

$XY = \frac{1}{2}$ PR .....(ii)

$ZY = \frac{1}{2}$ PQ .....(iii)

Adding the equation (i), (ii) and (iii)

$ZX + XY + ZY = \frac{1}{2}$ (QR + PQ + PR)

$\frac{1}{2} \times 24 = 12$ cm

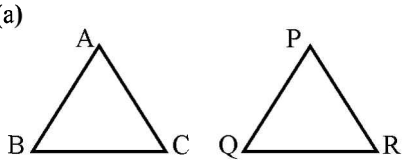
So circumference of ΔXYZ = 12 cm

4. ΔABC, and ΔPQR are similar and their perimeters are 36 and 24 respectively. If PQ = 10 then find the value of AB.

(a) 15 (b) 16
(c) 20 (d) 18

RRB JE - 26/06/2019 (Shift-I)

Ans : (a)



∵ ΔABC ~ ΔPQR

∴ $\frac{\text{Perimeter of } \Delta ABC}{\text{Perimeter of } \Delta PQR} = \frac{AB}{PQ}$

$\frac{36}{24} = \frac{AB}{10}$

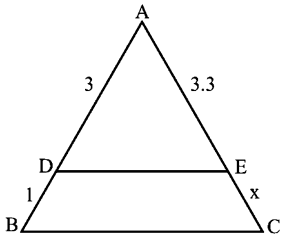
$AB = \frac{360}{24} = 15$

5. In ΔABC, the points D and E are on the sides AB and AC respectively such that DE ∥ BC and AD : DB = 3 : 1. If EA = 3.3 cm, find the value of AC.

(a) 5.5 cm (b) 4 cm
(c) 4.4 cm (d) 1.1 cm

RRB RPF Constable – 25/01/2019 (Shift-I)

Ans : (c)



from the thale'ss theorem-

$\frac{AD}{AB} = \frac{AE}{EC}$

$\frac{3}{1} = \frac{3.3}{EC}$

EC = 1.1

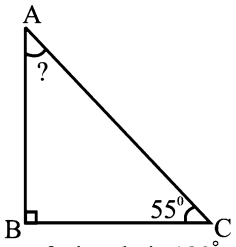
AC = AE + EC = 3.3 + 1.1 = 4.4

6. If acute angle of a right triangle is 55°, what will be the value of the other acute angle?

(a) 25° (b) 30°
(c) 40° (d) 35°

RRB Group-D – 12/10/2018 (Shift-III)

Ans : (d)



Sum total of angles of triangle is 180°

180° = ∠A+∠B+∠C

180°= ∠A + 90° + 55°

180°–145° = ∠A

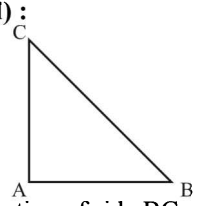
∠A = 35°

7. ABC is a right triangle whose angle A is right angle. Which side needs to bisect for the formation of two other right-angled triangles?

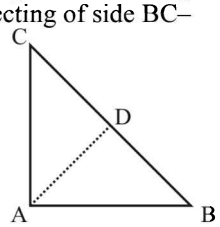
(a) AB (b) CA
(c) at the center (d) BC

RRB Group-D – 20/09/2018 (Shift-I)

Ans. (d) :



On bisecting of side BC–



Right Triangle = Δ ADB, Δ ADC

ABC is right triangle which angle A right angle, for the formation of two other right triangle, side BC will be bisected.

8. Which of the following are sides of a right angled triangle?

(a) 84 cm, 13 cm, 85 cm
(b) 84 cm, 63 cm, 115 cm
(c) 15 cm, 112 cm, 111 cm
(d) 76 cm, 100 cm, 57 cm

RRB Group-D – 25/09/2018 (Shift-III)

Ans. (a) : from pythagoras theorem

In right triangle, the square of hypotenuse is equal to the square of other two sides.

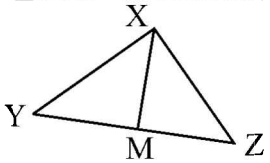
$\therefore (85)^2 = (13)^2 + (84)^2$

7225 = 169 + 7056

7225 = 7225

Sides of right angle Δ will be 85, 84 and 13

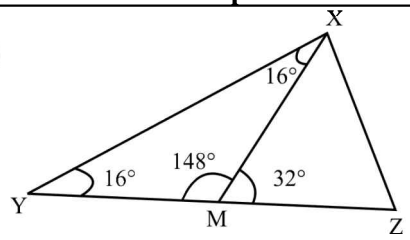
9. In the diagram M is mid point of $\overline{YZ}$, ∠XMZ = 32° and ∠XYZ = 16°. The value of ∠XZY is–



(a) 84° (b) 81°
(c) 74° (d) 68°

RRB Group-D – 27/09/2018 (Shift-I)

Ans. (c)



$\angle XMY = 180° - 32° = 148°$

Then $\angle YXM = 180 - (148 + 16) = 16°$

So $\angle MXZ = 90 - 16 = 74°$

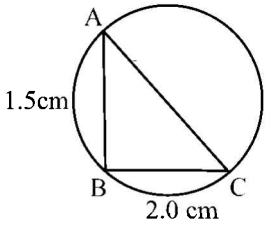
Then $\angle XZY = 180 - (32 + 74) = 74°$

10. Two adjacent sides of a right angled triangle are 1.5 cm and 2 cm long. The area of its circumcircle will be–

(a) 1.75π cm^2 (b) 1.25π cm^2
(c) 1.5625π cm^2 (d) 1.5π cm^2

RRB Group-D – 18/09/2018 (Shift-III)

Ans. (c) :



$(\text{hypotenuse})^2 = (\text{base})^2 + (\text{parpendicular})^2$

$(AC)^2 = (1.5)^2 + (2)^2$

$(AC)^2 = 2.25 + 4$

$(AC)^2 = 6.25$

root $= \sqrt{6.25} = 2.5\text{cm}$

∵ The radius of circum circle is half of hypotenuse.

$R = \frac{AC}{2} = \frac{2.5}{2} = \frac{5}{4}$

area of circum circle $= \pi R^2$

$$= \left(\frac{5}{4}\right)^2 \times \pi$$

$$= \frac{25}{16} \times \pi$$

$$= 1.5625\pi \text{ cm}^2$$

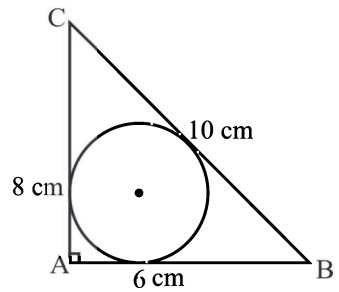
11. ABC is a right angled triangle in which ∠A = 90°, AB = 6 cm and AC = 8 cm. What will be perimeter of a circle whose centre O is inside the triangle?

(a) 4.5 π (b) 3π

(c) 3.5π (d) 4π

RRB Group-D – 26/10/2018 (Shift-III)

Ans : (d)



area of triangle $= \frac{1}{2} \times 6 \times 8 = 24\text{cm}^2$

$$S = \frac{a+b+c}{2} = \frac{8+6+10}{2} = 12$$

radius of incircle (r) = $\frac{\text{area of triangle}(\Delta)}{\text{Semiperimeter of triangle}(S)}$

$$r = \frac{\Delta}{S} = \frac{24}{12}$$

$\boxed{r = 2}$

Perimeter of circle $= 2\pi r$

$= 2 \times \pi \times 2$

$= 4\pi$

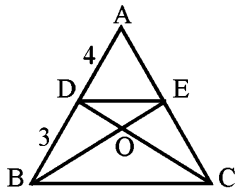
12. In triangle ABC, the points D and E lie on the sides AB and AC respectively. DE is parallel to the base BC. O is the intersection point of BE and CD. If AD : DB = 4:3, find the ratio of DO and DC.

(a) 4 : 11 (b) 3 : 7

(c) 5 : 12 (d) 5 : 7

RRB Paramedical Exam – 20/07/2018 (Shift-III)

Ans : (a)



DE ∥ BC

∴ Δ ADE ~ Δ ABC

$$\frac{AD}{AB} = \frac{DE}{BC}$$

$$\frac{4}{7} = \frac{DE}{BC} \quad (\because AB = AD + DB = 4 + 3 = 7)$$

from ΔDOE and ΔBOC

∠ODE = ∠ OCB (Alternate interior angle)

∠OED = ∠OBC (Alternate interior angle)

∠DOE = ∠BOC (Vertically opposite angle)

∴ Δ DOE ~ Δ COB

So $\frac{DO}{CO} = \frac{DE}{CB}$

$$\frac{DO}{CO} = \frac{4}{7}$$

or $\frac{CO}{DO} + 1 = \frac{7}{4} + 1$

$$\frac{DC}{DO} = \frac{11}{4}$$

or DO : DC = 4 : 11

13. Triangle ABC is similar to triangle PQR. If AB = 5 cm and PQ = 3 cm, then find the value of $\frac{ar(\Delta ABC)}{ar(\Delta PQR)}$.

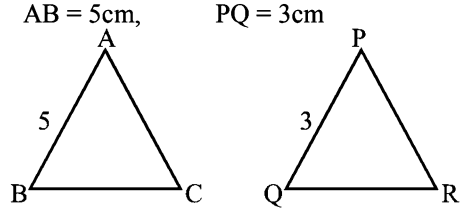
(a) $\frac{9}{25}$ (b) $\frac{3}{5}$

(c) $\frac{5}{3}$ (d) $\frac{25}{9}$

RRB Group-D – 19/09/2018 (Shift-III)

Ans. (d) : ΔABC ≅ ΔPQR

AB = 5cm, PQ = 3cm



The ratio of area of two similar triangles is equal to the ratio of their corresponding sides

$$\frac{\text{area}(\Delta ABC)}{\text{area}(\Delta PQR)} = \left(\frac{5}{3}\right)^2 = \frac{25}{9}$$

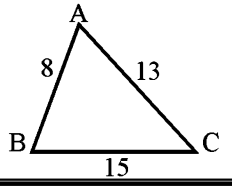
14. The lengths of the three sides of a triangle are 8 cm, 13 cm and 15 cm respectively. What will be the ratio of their altitudes?

(a) 195 : 120 : 104 (b) 15 : 13 : 8

(c) 28 : 23 : 21 (d) 104 : 195 : 120

RRB Group-D – 22/10/2018 (Shift-III)

Ans : (a)



Ratio of altitudes

$$= \frac{1}{8} : \frac{1}{13} : \frac{1}{15}$$

$$= 15\times13 : 8 \times 15 : 13 \times 8$$

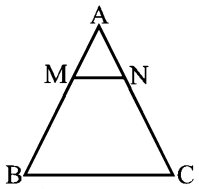
$$= 195 : 120 : 104$$

15. In triangle ABC, point M is on side AB and point N is on side AC such that BMNC becomes a trapezium. The ratio of side MN and side BC is 7:9. Find the ratio of the area of triangle AMN and the area of trapezium BMNC.

(a) 7 : 9 (b) 32 : 49
(c) 49 : 32 (d) 49 : 81

RRB Group-D – 11/12/2018 (Shift-II)

Ans : (c)



$\because$ BMNC is a trapezium

$\therefore$ MN || BC

$\therefore$ $\Delta AMN \sim \Delta ABC$

So $\frac{\text{area of } \Delta AMN}{\text{area of } \Delta ABC} = \left(\frac{MN}{BC}\right)^2 = \left(\frac{7}{9}\right)^2 = \frac{49}{81}$

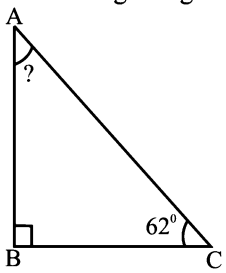
$\therefore \frac{\text{area of } \Delta AMN}{\text{area of } \Delta BMNC} = \frac{49}{81 - 49} = \frac{49}{32}$

16. The acute angle of a right angled triangle is 62°. Find the value of the second acute angle.

(a) 38° (b) 28°
(c) 45° (d) 36°

RRB Group-D – 15/10/2018 (Shift-III)

Ans. (b) : Given right angle triangle



$\angle B = 90°$

$\angle C = 62°$

$\angle A = ?$

Sum of interior angles of triangle is 180°

$\angle A = 180° - (90° + 62°)$

$= 180° - 152° = 28°$

17. If 8, p and 17 are pythagoras triplet, then the value of p will be–

(a) 15 (b) 9
(c) 14 (d) 13

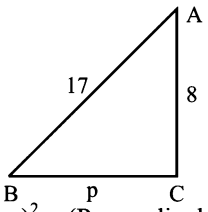
RRB Group-D – 01/10/2018 (Shift-I)

Ans. (a) : According to the question,

8, p and 17 is Pythagoras triplet

then p = ?

from the figure,



$(\text{hypotenuse})^2 = (\text{Perpendicular})^2 + (\text{base})^2$

$(17)^2 = (8)^2 + (p)^2$

$289 = 64 + (p)^2$

$289 - 64 = (p)^2$

$225 = p^2$

$\boxed{p = 15}$

18. The area of a triangle ABC is 63 square unit. Two parallel lines DE, FG are drawn such that it divides AB and AC into three equal parts. What is the area of quadrilateral DEFG?

(a) 28 Square unit (b) 35 Square unit
(c) 21 Square unit (d) 48 Square unit

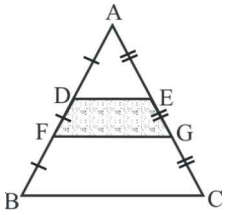
RRB NTPC 02.04.2016 Shift : 2

Ans : (c) in similar Δ ADE and ABC–

$$\frac{\text{area of } \Delta ADE}{\text{area of } \Delta ABC} = \left(\frac{AD}{AB}\right)^2$$

$$\frac{\text{area of } \Delta ADE}{63} = \left(\frac{AD}{3AD}\right)^2$$

area of Δ ADE = 7 square unit



$\Rightarrow$ $\Delta AFG \cong \Delta ABC$

$\therefore$ $\frac{\text{are of } \Delta AFG}{\text{area of } \Delta ABC} = \left(\frac{AF}{AB}\right)^2$

$\Rightarrow$ $\frac{\text{area of } \Delta AFG}{63} = \left(\frac{2AD}{3AD}\right)^2$

$\Rightarrow$ area of Δ AFG $= \frac{4}{9} \times 63 = 28$ Square unit

$\therefore$ area of Δ DEFG = ΔAFG – Δ ADE

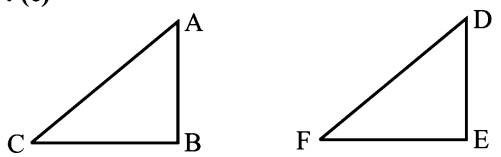
= 28 – 7 = 21 Square unit

19. If ΔABC and ΔDEF are similar triangles, in which BC = 4 cm, EF = 7 cm and the area of ΔABC is 144 square cm, then find the area of ΔDEF.

(a) 252 Square cm (b) 504 Square cm
(c) 441 Square cm (d) 324 Square cm

RRB NTPC 02.04.2016 Shift : 1

Ans : (c)



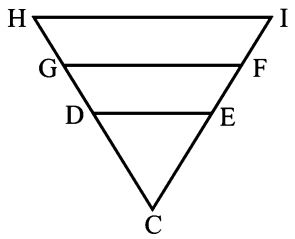
$\because \Delta ABC \sim \Delta DEF$

$\therefore \dfrac{\text{area of } \Delta ABC}{\text{area of } \Delta DEF} = \left(\dfrac{BC}{EF}\right)^2$

$\dfrac{144}{\text{area of } \Delta DEF} = \left(\dfrac{4}{7}\right)^2$

area of $\Delta DEF = \dfrac{144 \times 49}{16} = 441$ Square cm

20.

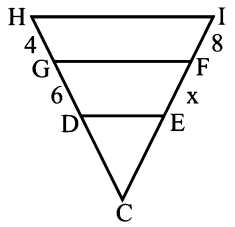


HI, GF and DE are parallel lines, If DG = 6, GH = 4 and FI = 8, then EF = ?

(a) 8 (b) 9
(c) 12 (d) 16

RRB NTPC 31.03.2016 Shift : 1

Ans : (c)



from the thales's theorem
in ΔCGF,
Let EF = x

$\dfrac{CD}{GD} = \dfrac{CE}{EF}$

or $\dfrac{CD}{CE} = \dfrac{GD}{EF}$ .....(1)

again in ΔCHI,

$\dfrac{CD}{DH} = \dfrac{CE}{EI}$

$\dfrac{CD}{CE} = \dfrac{DH}{EI}$ .....(2)

$\therefore$ from the equation (1) and (2)

$\dfrac{GD}{EF} = \dfrac{DH}{EI}$

$\dfrac{6}{x} = \dfrac{6+4}{x+8}$

$10x = 6x + 48$

$4x = 48$

$x = 12$

EF = 12

21. If the ratio of the angles of a triangle is 1:2:3, find the value of the largest angle?

(a) 30° (b) 60°
(c) 90° (d) 120°

RRB NTPC 19.04.2016 Shift : 1

Ans : (c) If angles of triangle are x, 2x and 3x respectively

Then $x + 2x + 3x = 180°$

$6x = 180°$

$x = 30°$

$\therefore$ largest angle $= 3x = 3 \times 30 = 90°$

22. If $\Delta ABC \cong \Delta XYZ$ and angle BAC = 55°, then angle ZXY = ?

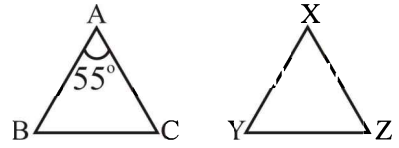
(a) 65° (b) 135°
(c) 55° (d) 67.5°

RRB ALP & Tec. (31-08-18 Shift-I)

Ans : (c) given− $\angle BAC = 55°$

$\Delta ABC \cong \Delta XYZ$

So from congruence rules−



$\angle A = \angle X$

or $\angle BAC = \angle ZXY$

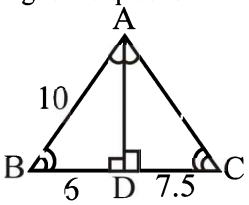
$55° = \angle ZXY$

23. In ΔABC, AB = 10 cm. BD = 6 cm and DC = 7.5 cm. $\angle A$ is bisected internally to intersect BC at D. What is the length of CA?

(a) 12 cm (b) 10 cm
(c) 10.5 cm (d) 12.5 cm

RRB ALP & Tec. (30-08-18 Shift-III)

Ans : (d) according to the question in ΔABC



$\Delta ADB \cong \Delta ADC$

By the law of similarty,

$\dfrac{AB}{DB} = \dfrac{AC}{DC}$

$\dfrac{10}{6} = \dfrac{AC}{7.5}$, $AC = \dfrac{10 \times 7.5}{6}$

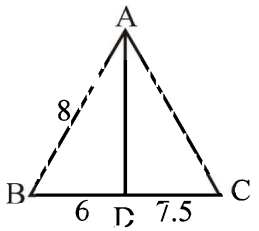
AC = 12.5 cm

24. In ΔABC, AB = 8 cm. The bisector of ∠A is internally meets on BC at D. and BD = 6 cm and DC = 7.5 cm. What will be the value of CA?

(a) 10.5 cm (b) 12.5 cm
(c) 12 cm (d) 10 cm

RRB ALP & Tec. (10-08-18 Shift-III)

Ans : (d)



from the interior angle Bisector theorem–

$$\frac{AB}{BD}=\frac{AC}{CD}$$

$$\frac{8}{6}=\frac{AC}{7.5}$$

$$AC=\frac{8\times 7.5}{6}$$

AC = 10 cm

25. In ΔABC, D and E are points on AB and AC respectively in which DE is parallel to BC. If AD = 2 cm, BD = 3 cm then what is the value of $\frac{ar(\Delta ADE)}{ar(\Delta ABC)}$?

(a) $\frac{16}{81}$ (b) $\frac{4}{9}$
(c) $\frac{4}{25}$ (d) $\frac{2}{5}$

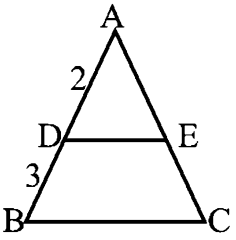
RRB Group-D – 22/09/2018 (Shift-III)

Ans. (c) : ΔABC in

∵ DE//BC ∴ ΔADE ~ ΔABC

AD = 2 cm

BD = 3 cm



in similar ΔADE and ΔABC

$$\frac{ar(\Delta ADE)}{ar(\Delta ABC)}=\frac{AD^2}{AB^2}$$

$$=\frac{(2)^2}{(2+3)^2}=\frac{4}{25}$$

Type - 2

26. All four angles of a quadrilateral are the same. Find their values?

(a) 110° (b) 80°
(c) 75° (d) 90°

RRB Group-D – 09/10/2018 (Shift-I)
RRB RPF SI – 12/01/2019 (Shift-I)

Ans. (d) : Sum of four angles of quadrilateral = 360°

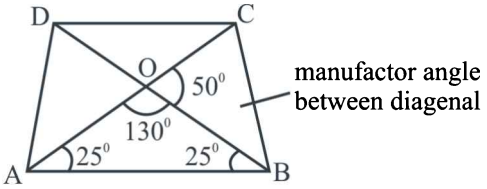
$$\therefore \text{ each angle}=\frac{360^\circ}{4}=90^\circ$$

27. The diagonals of a quadrilateral are inclined 25° on one side of the quadrilateral. The acute angle to be drawn between the diagonals will be–

(a) 40° (b) 50°
(c) 55° (d) 25°

RRB Group-D – 24/09/2018 (Shift-I)

Ans : (b)



in ΔAOB

∠A + ∠B + ∠O = 180°

25° + 25° + ∠O = 180°

obtuse angle (∠O) = 180°–50° = 130°

acute angle (∠COB) = 180°–130°

= 50°

So acute angle formed between diagonals will be 50°

28. The ratio of four angles of a quadrilateral is 3:4:5:6. What will be the value of smallest angle?

(a) 49° (b) 60°
(c) 45° (d) 36°

RRB Group-D – 05/10/2018 (Shift-I)

Ans. (b) : Let four interior angles of quadrilateral are 3x, 4x, 5x, 6x respectively

Sum of four interior angles of quadrilater al = 360°

3x+4x+5x+6x= 360°

18x= 360°

x = 20°

So smallest angle = 3×20

= 60°

29. The values of two adjacent angles of a quadrilateral are 125° and 35° and the other two angles are equal. Find the value of equal angles.

(a) 90° (b) 100°
(c) 135° (d) 80°

RRB Group-D – 22/10/2018 (Shift-II)

Ans : (b) ∵ Sum of all interior angles of quadrilatiral = 360°

Two angles of quadrilaterals are equal

let each of them = x°

∴ 125° + 35° + x° + x° = 360°

2x° = 360° – 160

x° = 100

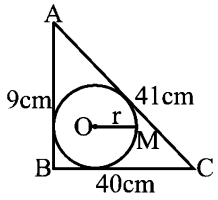
30. What is the radius of a circle formed inside a triangle with sides 9 cm, 40 cm and 41 cm?

(a) 3.5 cm (b) 3 cm

(c) 2.5 cm (d) 4 cm

RRB Group-D – 03/12/2018 (Shift-II)

Ans : (d) A right-angled triangle is formed from the sides of the measure of 9 cm, 40 cm and 41 cm.



Then area of $\Delta = \frac{1}{2} \times 9 \times 40$

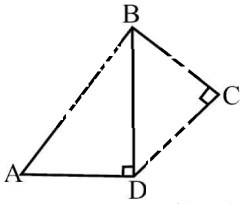
$= 180 cm^2$

Semi perimeter $(s) = \frac{1}{2}[40 + 41 + 9]$

= 45 cm

So radius of in circle $(r) = \frac{\Delta}{s} = \frac{180}{45} = 4cm$

31. In the given figure ∠BAD = 30⁰, ∠DBC = 60⁰ If the length of $\overline{AB}$ is 18 cm, find the length of $\overline{BC}$ –

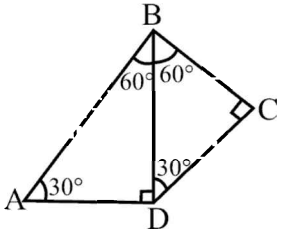


(a) 7.5cm (b) 4.5cm

(c) 9cm (d) 6cm

RRB Group-D – 12/12/2018 (Shift-I)

Ans. (b)



from ΔABD,

$\frac{AB}{\sin 90°} = \frac{BD}{\sin 30°}$

$BD = \frac{AB}{2} = \frac{18}{2}$

BD = 9 cm

∵ from ΔBDC,

$\frac{BD}{\sin 90°} = \frac{BC}{\sin 30°}$

$BC = \frac{1}{2} \times BD = \frac{1}{2} \times 9$

BC = 4.5 cm

32. The value of each of the three acute angles of a quadrilateral is 82°. Find the value of the fourth angle.

(a) 125° (b) 114°

(c) 120° (d) 100°

RRB Group-D – 25/10/2018 (Shift-II)

Ans : (b) Let value of fourth angle = x°

∵ Sum of four interior angles of quadrilateral = 360°

∴ x + 82° + 82° + 82° = 360°

x = 360° - 246° = 114°

33. Three angles of a quadrilateral are same, and the value of the fourth angle is 150°. What is the value of each of the same angles?

(a) 75° (b) 90°

(c) 80° (d) 70°

RRB Group-D – 08/10/2018 (Shift-I)

Ans. (d) : Sum of four angles of quadrilateral is 360°

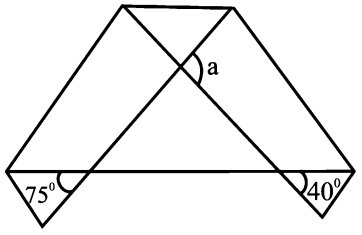
given-

fourth angle = 150°

Sum of other three angles = 360°–150° = 210°

So value of each angle = $\frac{210°}{3} = 70°$

34. What is the value of a?

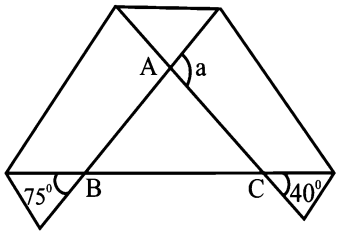


(a) 65° (b) 75°

(c) 105° (d) 115°

RRB NTPC 29.03.2016 Shift : 3

Ans : (d)



$\angle ABC = 75°$ (opposite angle)

$\angle ACB = 40°$ (opposite angle)

$\therefore \angle BAC = 180° - (75° + 40°)$

$= 180° - 115° = 65°$

but $\angle BAC + a = 180°$ (angle of Linear pair)

$a = 180° - 65° = 115°$

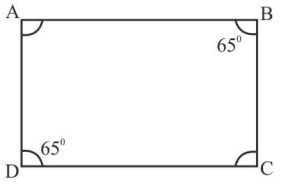
35. The value of each of the two angles of a quadrilateral is 65°. The other two angles are also equal. Find the value of these angles.

(a) 100^0 (b) 115^0

(c) 125^0 (d) 105^0

RRB Group-D – 03/10/2018 (Shift-II)

Ans : (b) Sum of four angles of any quadrilateral is 360°



$\angle B = \angle D = 65^0$

$\angle A = \angle C = x^0$

Then $65^0 + 65^0 + x^0 + x^0 = 360$

$2x^0 = 360–130$

$\boxed{x = 115^0}$

So value of remain each angle $= 115^0$

Type - 3

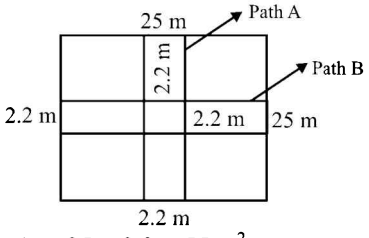
36. Each side of a quadrate park is 25 cm and has 2.2 m wide two paths passing through its centre. How much rupees will its cost to spread concrete on the paths at a cost of ₹1/dm^2?

(a) ₹ 11,000 (b) ₹ 110

(c) ₹ 105.16 (d) ₹ 10,516

RRB Group-D – 01/10/2018 (Shift-II)

Ans. (d) :



area of path A $= 25 \times 2.2 = 55\ m^2$

area of path B $= 25 \times 2.2 = 55\ m^2$

Total area of street $= 25 \times 2.2 + 25 \times 2.2 - 2.2 \times 2.2$

$= 55 + 55 - 4.84 = 105.16\ m^2$

Total cost $= 105.16 \times 100 =$ ₹ 10516

37. In which all the sides have same length?

(a) Rectangle (b) Trapezium

(c) Parallelogram (d) Square

RRB NTPC 30.04.2016 Shift : 1

Ans : (d) All four sides are the same length in both square and rhombus. but the angle of each of the four corners in the square is right-angled, while the rhombus face to face is the same.

Type - 4

38. The length of a rectangular plot is 5 m more than its width. If the circumference of the plot is 142 m, find the dimensions of the plot.

(a) Length 38 m and width 33 m

(b) Length 39 m and width 34 m

(c) Length 34 m and width 39 m

(d) Length 33 m and width 38 m

RRB RPF Constable – 19/01/2019 (Shift-II)

Ans. (a) :

Suppose the length of rectangular plot is l meter and breadth b meter.

$\because$ Circumference of rectangular plot $= 2(\ell + b)$

According to the question,

$\because \quad \ell = b + 5$

$\therefore \quad 2(\ell + b) = 142$

$\Rightarrow \quad 2(b + 5 + b) = 142$

$\Rightarrow \quad (2b + 5) = 71$

$\Rightarrow \quad 2b = 71 - 5$

$\Rightarrow \quad 2b = 66$

$\Rightarrow \quad b = 33m$

$\ell = 33 + 5 = 38m$

So length will be 38 meter and breadth 33 meter.

39. If a rectangle length : width = 4:3. Find the value of length : diagonal.

(a) 4 : 7 (b) 2 : 3

(c) 4 : 5 (d) 1 : 5

RRB JE - 24/05/2019 (Shift-I)

Ans : (c) Let the length of rectangle $= 4x$

and breadth of rectangle $= 3x$

$\therefore$ diagonal of rectangle $= \sqrt{(4x)^2 + (3x)^2} = 5x$

$\therefore$ length : diagonal $= 4x : 5x = 4 : 5$

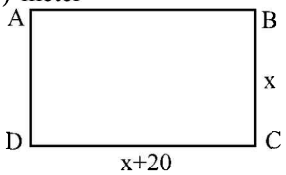
40. The length of a rectangular plot is 20 m more than its width. If the cost of planting the fence at ₹26.50 per meter on the plot is ₹ 5300, find the length of the plot in meters.

(a) 50 m (b) 40 m

(c) 120 m (d) 60 m

RRB JE - 29/05/2019 (Shift-I)

Ans : (d) Let width of rectangular plot = x m
length of plot = (x + 20) m
Perimeter = 2 (length + width)
= 2 (x + x + 20) meter
= (4x + 40) meter



According to the question
$(4x + 40) \times 26.50 = 5300$
$4x + 40 = 200$
$4x = 160$
x = 40 meter
length of plot = (x + 20) meter = (40 + 20) meter
= 60 meter

41. The ratio of length and width of a rectangle is 3:1. If its perimeter is 96 m, then what is the length of rectangle?

(a) 48 m (b) 24 m
(c) 36 m (d) 12 m

RRB Group-D – 17/09/2018 (Shift-II)

Ans : (c) Let length of rectangle is 3x m. and breadth is x m
Perimeter of rectangle (P) = 2 [l + b]

$$\begin{bmatrix} l = \text{lendth} \\ b = \text{breadth} \end{bmatrix}$$

$96 = 2 [3x + x]$
$96 = 6x + 2x$
$96 = 8x$
$x = \frac{96}{8}$
$\boxed{x = 12}$
length of rectangle = 3 × 12
= 36 meter

42. The dimensions of the rectangular plot of land are given in the nearest integer, 37 m and 23m. What is the upper limit of the circumference of the rectangle?

(a) 122 m (b) 61 m
(c) 60 m (d) 120 m

RRB Group-D – 12/10/2018 (Shift-I)

Ans. (a) : given-
length of rectangle = 37 m
breadth of rectangle = 23 m
circumference of rectangle
= 2 (length + breadth)
= 2 (37 + 23)
= 120 meter
So this answer will be 122 m. Because the nearest integer is given here and asked about the upper limit.

Type - 5

43. The length of one side of a rhombus is 12 cm and the length of one of its diagonals is also 12 cm. The length of second diagonal will be _ cm.

(a) $12\sqrt{3}$ (b) $6\sqrt{3}$
(c) 24 (d) $9\sqrt{3}$

RRB RPF SI – 10/01/2019 (Shift-I)

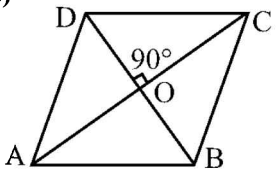
Ans : (a) Let second diagonal of rhombus is d_2
side (a) of rhombus $= \frac{1}{2}\sqrt{d_1^2 + d_2^2}$
where d_1 & d_2 is diagonal of rhombus
$12 = \frac{1}{2}\sqrt{12^2 + d_2^2}$
$12 \times 2 = \sqrt{12^2 + d^2}$
Square on both sides
$(24)^2 = 12^2 + d_2^2$
$d_2^2 = (24)^2 - (12)^2$
$= (24 + 12)(24 - 12)$
$= 36 \times 12 = 432$
$\therefore d_2 = 12\sqrt{3}$cm.

44. The length of one side of a rhombus is $\frac{17}{3}$ cm and its one diagonal is $\frac{16}{3}$ cm. Find the length of second diagonal.

(a) $\frac{20}{3}$ cm (b) $\frac{32}{3}$ cm
(c) 10 cm (d) $\frac{16}{3}$cm

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (c)



side of rhombus $= \frac{17}{3}$cm
first diagonal $AC = \frac{16}{3}$cm
$\because$ AO = OC and BO = OD
$\therefore AO = OC = \frac{AC}{2}$
$= \frac{16}{3} \Big/ 2$
$AO = \frac{8}{3}$cm
ΔAOB in, $AB^2 = AO^2 + BO^2$

$BO = \sqrt{AB^2 - AO^2}$

$= \sqrt{\left(\frac{17}{3}\right)^2 - \left(\frac{8}{3}\right)^2}$

$= \sqrt{\frac{289}{9} - \frac{64}{9}}$

$= \sqrt{\frac{225}{9}}$

$= \frac{15}{3}$

$BO = 5$

So $OB = OD = 5$ cm

Second diagonal $BD = BO + OD = 5 + 5 = 10$ cm

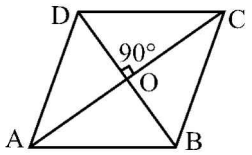
45. The length of the side of rhombus is 5 cm and it's area is 4 cm^2. What will be the sum of the lengths of its diagonals?

(a) 5 cm (b) 6 cm

(c) 8 cm (d) 7 cm

RRB Group-D – 11/10/2018 (Shift-II)

Ans : (b)



$AB = \sqrt{5}$

Rhombus

In ABCD,

first diagonal = AC

second diagonal = BD

$4AB^2 = AC^2 + BD^2$

$4 \times 5 = AC^2 + BD^2$

$20 = AC^2 + BD^2$

Let $AC = x$

$BD = y$

$\therefore$ area of rhombus $= \frac{1}{2} \times x \times y = 4$

$x \times y = 8$

$(x+y)^2 = 20 + 2\times8$

$(x+y)^2 = 20 + 16$

$(x+y)^2 = 36$

$\boxed{x + y = 6 \text{ cm}}$

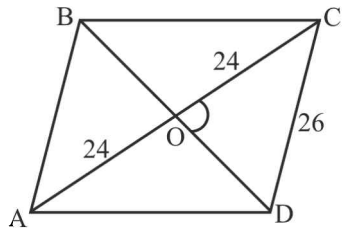
46. The length of one side of a rhombus is 26 cm and the length of one diagonal is 48 cm, then what will be the length of the second diagonal?

(a) 24 cm (b) 20 cm

(c) 25 cm (d) 22 cm

RRB Group-D – 10/10/2018 (Shift-I)

Ans : (b)



$(OD)^2 = (CD)^2 - (OC)^2$ (from pythagoras theorem)

$(OD)^2 = (26)^2 - (24)^2$

$(OD)^2 = 676 - 576$

$(OD)^2 = 100$

$OD = 10$ cm

diagonal $BD = OB + OD$ $\{\therefore OB = OD\}$

$BD = 20$ cm

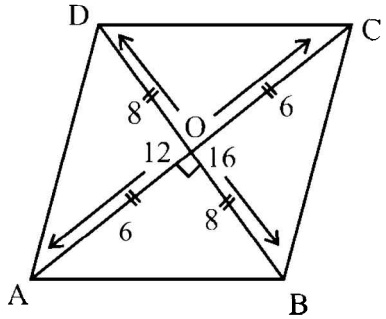
47. The length of diagonals of a rhombus are 16 cm and 12 cm. What is the perimeter of the rhombus?

(a) 56 cm (b) 20 cm

(c) 28 cm (d) 40 cm

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (d) : In the rhombus $d_1 = 16$, $d_2 = 12$ perimeter = ?



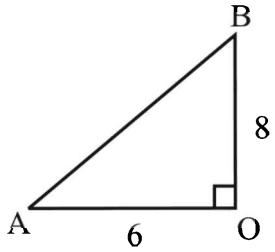
diagonals of rhombus bisect each other at right angles.

$AO = OC$

$BO = OD$

$\angle AOB = \angle BOC = \angle COD = \angle DOA = 90°$

sides AB of rhombus



$AB^2 = AO^2 + BO^2$

$AB^2 = 6^2 + 8^2$

$AB^2 = 100$

$AB = 10$

Perimeter of rhombus $= 4a$

$= 4 \times 10$

$= 40$ cm

48. The length of one side of a rhombus is 4 cm and length of one diagonal is 6 cm. The length of second diagonal wil be ___cm.

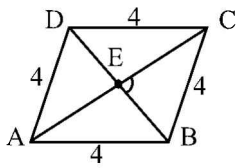
(a) $\sqrt{14}$ (b) $\frac{\sqrt{14}}{2}$

(c) $\sqrt{7}$ (d) $2\sqrt{7}$

RRB Group-D – 16/11/2018 (Shift-I)

Ans. (d) In rhombus, all sides are equal and diagonals of rhombus bisect each other at right angles.

in ΔBEC,



$BC^2 = CE^2 + EB^2$ $\left[BE = \frac{BD}{2} = 3\right]$

$4^2 = CE^2 + 3^2$

$CE = \sqrt{7}$

$CA = 2 \times CE$

$CA = 2 \times \sqrt{7}$

$CA = 2\sqrt{7}$

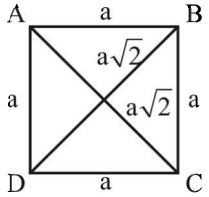
49. The sum of the diagonals of a square is ___ the perimeter of the square.

(a) less than $\frac{3}{4}$ of (b) larger than

(c) equal to (d) less than $\frac{1}{2}$ of

RRB Group-D – 16/11/2018 (Shift-III)

Ans. (a) :



Sum of diagonals $a\sqrt{2} + a\sqrt{2} = 2a\sqrt{2}$

Perimeter $= a + a + a + a = 4a$

$\frac{3}{4}$ times of perimeter $= 4a \times \frac{3}{4} = 3a$

$\Rightarrow 2a\sqrt{2} < 3a$

Thus, the sum of the diagonals of a square is less than $\frac{3}{4}$ the times of the perimeter of the square.

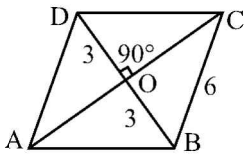
50. The length of one side of rhombus and one of the two diagonals is 6 m. The area of rhombus is _____ cm^2.

(a) 18 (b) $9\sqrt{3}$

(c) $27\sqrt{3}$ (d) $18\sqrt{3}$

RRB Group-D – 18/09/2018 (Shift-I)

Ans. (d) :



According to the figure,

in ΔOBC

$(OC)^2 = (BC)^2 - (OB)^2$

$(OC)^2 = (6)^2 - (3)^2$ (from phythagoras theorem)

$(OC) = 3\sqrt{3}$

area of $\Delta OBC = \frac{1}{2} \times 3 \times 3\sqrt{3}$

$= \frac{9\sqrt{3}}{2} cm^2$

area of rhombus ABCD = 4x ΔOBC area

$= 4 \times \frac{9\sqrt{3}}{2} cm^2$

$= 18\sqrt{3} cm^2$

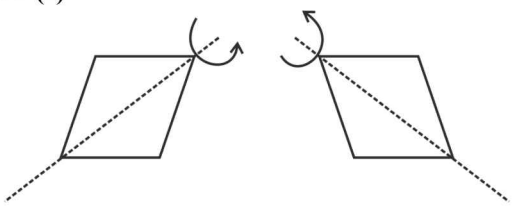
51. The order of rotational symmetry of a rhombus is–

(a) 1 (b) 4

(c) 2 (d) 0

RRB NTPC 22.04.2016 Shift : 2

Ans : (c)



Hence the rotational symmetry will be 2.

Type - 6

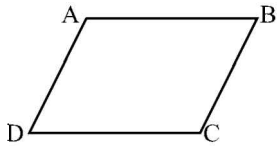
52. In a parallelogram ABCD, ∠A = (3x-25)° and ∠C = (2x+15)°, In which angle A and angle C facing angle. Find the angle A–

(a) 105° (b) 85°

(c) 95° (d) 115°

RRB RPF SI – 13/01/2019 (Shift-II)

Ans : (c)



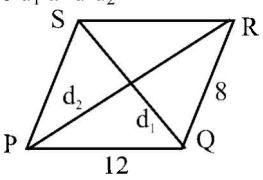
$\angle A = (3x–25)°$
$\angle C = (2x+15)°$
$\angle A = \angle C$ [opposite angle is equal]
$(3x – 25)° = (2x+15)°$
$x = 40°$
$\angle A = (3x – 25)° = 120 – 25 = 95°$

53. A parallel quadrilateral PQRS with lengths of sides 8 cm and 12 cm has a diagonal 10 cm long. The second diagonal is approximately.

(a) 18 cm (b) 17.8 cm
(c) 17 cm (d) 17.5 cm

RRB Group-D – 17/09/2018 (Shift-II)

Ans : (b) If the adjacent sides of a parallelogram are a and b and the diagonals are d_1 and d_2 -



$d_1^2 + d_2^2 = 2\left(a^2 + b^2\right)$

given, a = 12cm, b = 8cm, d_1 = 10cm

$\therefore d_2^2 = 2\left(12^2 + 8^2\right) - 10^2$

$= 2 \times (144 + 64) – 100$

$= 2 \times 208 – 100$

$= 416 – 100 = 316$

$\therefore d_2 = \sqrt{316} = 17.8$ cm

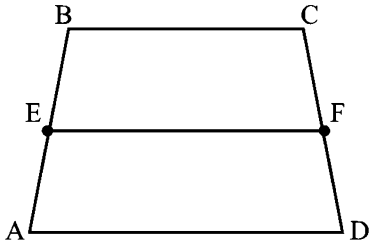
54. If in a asymmetric ABCD, E and F are the midpoints of the two transverse sides AB and CD, then FE = ?

(a) $\frac{2}{3}(AB – CD)$ (b) $\frac{1}{2}(AB + CD)$

(c) $\frac{\sqrt{3}}{2}(AB + CD)$ (d) $\sqrt{3}(AB + CD)$

RRB Group-D – 19/09/2018 (Shift-III)

Ans. (b) : In any trapezium, the line joining the midpoint of the eassumetric sides is parallel to the parallel sides of the trapezium and half of their sum in length.



$FE = \frac{1}{2}(AB + CD)$

55. The smallest side of a parallelogram is 4.8 cm and the largest side is 1.5 times to the smallest side. Find the circumference of a parallelogram.

(a) 48 cm (b) 36 cm
(c) 24 cm (d) 72 cm

RRB Group-D – 23/10/2018 (Shift-II)

Ans. (c) : Smallest sides of parallelogram =4.8cm

largest sides of parallelogram = $4.8 \times 1.5 = 7.20$ cm

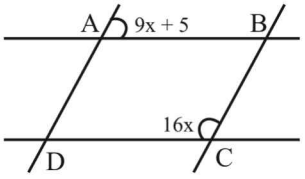
So circumference of Parallelogram = 2(small side + large side)

$= 2 \times (4.8 + 7.2)$

$= 2 \times 12.0$

= 24 cm

56. Find the value (in degrees) of angle ABC in the given parallelogram ABCD.



(a) 112° (b) 102°
(c) 78° (d) 68°

RRB Group-D – 11/10/2018 (Shift-I)

Ans : (d) $\angle D = 9x + 5$ (Corresponding angles)

then $9x + 5 + 16x = 180°$

($\therefore$ sum of adjacent angles is 180° in parallelogram)

$\Rightarrow 25x = 180° – 5°$

$\Rightarrow 25x = 175$

$\Rightarrow x = \frac{175}{25}$

$\Rightarrow x = 7$

then $16x = 16 \times 7 = 112°$

According to the question,

$16x + \angle ABC = 180°$

$\Rightarrow 112 + \angle ABC = 180°$

$\Rightarrow \angle ABC = 180° – 112$

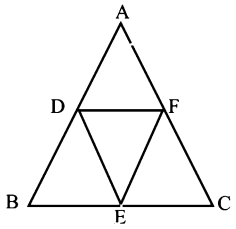
$\angle ABC = 68°$

57. In an equilateral triangle ABC, D, E, F are the midpoints of AB, BC and AC respectively. So the quadrilateral BEFD is:

(a) A square
(b) A rectangle
(c) A parallelogram
(d) A rhombus

RRB NTPC 29.03.2016 Shift : 1

Ans : (c)



In a triangle, the line joining the midpoints of two sides is Parallel and half of the third side.

DF || BC

and $DF = \frac{1}{2}BC$ .....(i)

$\therefore$ E is the mid point of BC

$\therefore$ $BE = \frac{1}{2}BC$ .....(ii)

from the equation (i) and equation (ii)

DF = BE

$\because$ DF || BC $\Rightarrow$ DF || BE

So quadrilateral BEFD is a parallelogram

58. The order of rotational symmetry of a parallelogram is:

(a) 1 (b) 4

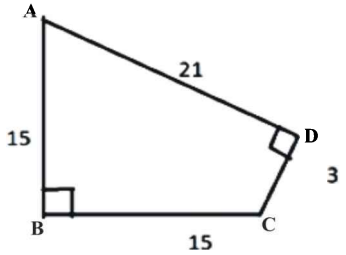
(c) 2 (d) 0

RRB NTPC 26.04.2016 Shift : 2

Ans : (c) The order of rotational symmetry of a Parallelogram is 2.

Type - 5

59. Find the area of the quadrilateral shown in the following figure (underlined is not measured).



(a) 144 (b) 54

(c) 63 (d) 123

RRB Group-D – 10/10/2018 (Shift-III)

Ans : (a)

area of quadrilaterals ABCD = area of ΔABC + area of ΔADC.

$\frac{1}{2}\times AB\times BC+\frac{1}{2}\times CD\times DA$

$\left(\frac{1}{2}\times 15\times 15+\frac{1}{2}\times 3\times 21\right)$

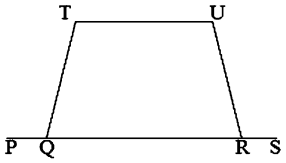
$=\frac{1}{2}(15\times 15+3\times 21)$

$=\frac{1}{2}(225+63)$

$=\frac{288}{2}$

$\boxed{=144}$

60. In the given diagram, TU||PS and points Q and R are located on PS also $\angle PQT = x°$, $\angle RQT = (x-50)°$ and $\angle TUR = (x+25)°$. What is the meassure of $\angle URS$



(a) 130° (b) 115°

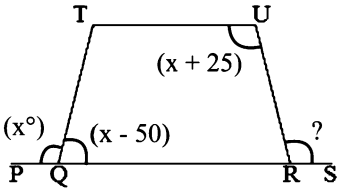
(c) 140° (d) 135°

RRB Group-D – 17/09/2018 (Shift-III)

Ans. (c) : given– TU||PS

$\angle PQT = x°, \angle RQT = (x-50)°$

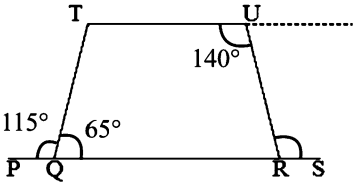
$\angle TUR = (x+25)°$ $\angle URS = ?$



x + x – 50 = 180

2x = 230

$\boxed{x = 115°}$ Putting value of x is diagram.



Alternate angle $\angle TUR = \angle URS$

$\angle URS = 140°$

61. Trapezium is a quadrilateral whose:

(a) All sides are equal.

(b) Opposite sides are equal.

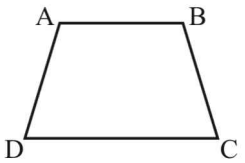
(c) Two pairs of parallel opposite sides are.

(d) One pair of parallel opposite sides is.

RRB NTPC 28.03.2016 Shift : 3

Ans : (d) Trapezium is a quadrilateral with a pair of parallel opposite sides.

AB || DC



62. The order of rotational symmetry of a trapezium is.

(a) 2 (b) 0

(c) 1 (d) 3

RRB NTPC 11.04.2016 Shift : 2

Ans : (c) the rotational symmetry of a trapezium is one order.

Note : If a shape is partially rotated, If the shape looks the same even after rotating, then this property is called rotary symmetry.

Type - 8

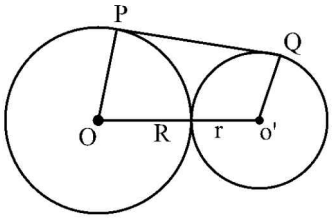
63. Two circles touch each other externally at point X. PQ is simple common tangent touching both circles at P and Q point. If the radius of circles are R and r, find the value of PQ^2.

(a) 4Rr (b) $2\pi Rr$

(c) $\frac{3}{2}$ Rr (d) 2Rr

RRB RPF Constable – 25/01/2019 (Shift-III)

Ans : (a)



$PQ = \sqrt{d^2 - (r_1 - r_2)^2}$

there d = O O' = (R + r)

$r_1 = R$

$r_2 = r$

then, $PQ = \sqrt{(R+r)^2 - (R-r)^2}$

$= \sqrt{R^2 + r^2 + 2Rr - R^2 - r^2 + 2Rr}$

$= \sqrt{4Rr}$

$PQ = 2\sqrt{Rr}$

$PQ^2 = \left(2\sqrt{Rr}\right)^2 = 4Rr$

64. The circumference of a circle is 132 cm. What will be the circumference of the part whose center angle is 135°?

(a) 93.5 cm (b) 101.5 cm

(c) 92.5 cm (d) 91.5 cm

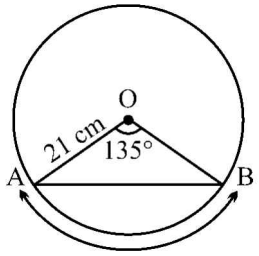
RRB JE - 26/05/2019 (Shift-II)

Ans : (d) According to the question

circumference of circle, $2\pi r = 132$

$2 \times \frac{22}{7} \times r = 132$

$r = \frac{132 \times 7}{2 \times 22} = 21$ cm



circumference $= 2\pi \frac{r\theta}{360} + 2r$

$= \frac{132 \times 135}{360} + 2 \times 21 = 49.5 + 42 = 91.5$ cm

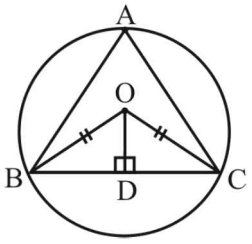
65. If 'O' is the circumcentre of triangle ABC and OD is perpendicular to BC, find the value of angle ∠BOD.

(a) (1/2) ∠A (b) 90° ∠A

(c) ∠A (d) 2∠A

RRB RPF SI – 16/01/2019 (Shift-III)

Ans : (c)



∵ The angle substended by the arc of a circle at the center is twice the angle subtended at the circumference by the same arc.

∠BOC = 2∠A

$\therefore \angle BOD = \frac{\angle BOC}{2} = \frac{2\angle A}{2} = \angle A$

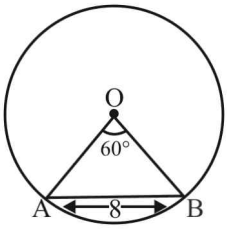
66. A chord of 8 cm length makes an angle of 60° at the center, what will be the value of radius of the circle?

(a) 8 cm (b) 6 cm

(c) 4 cm (d) 12 cm

RRB JE - 30/05/2019 (Shift-II)

Ans : (a)



According to the figure,

$\angle AOB = 60°$

$\because \quad \angle A = \angle B = 60°$

So ΔOAB will be equilateral triagnle

$\because$ OA = OB = AB = 8 cm

$\because$ OA and OB is circle's radius

So r = 8 cm

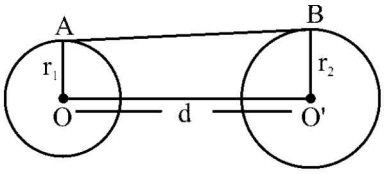
67. Radius r_1 and r_2 the distance between the centers of the two circles is d find the length of their tangent lines.

(a) $\sqrt{d^2\left(r_1^2 r_2^2\right)}$ (b) $\sqrt{d^2-\left(r_1-r_2\right)^2}$

(c) $\sqrt{d^2-\left(r_1^2 r_2^2\right)}$ (d) $\sqrt{d^2-\left(r_1+r_2\right)^2}$

RRB JE - 01/06/2019 (Shift-I)

Ans : (b)



length of tangent line AB

$AB = \sqrt{(\text{distance between centre's})^2 - (\text{difference of radius})^2}$

$= \sqrt{d^2-\left(r_1-r_2\right)^2}$

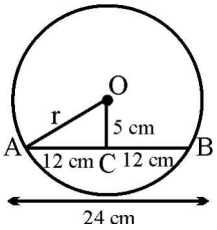
68. If there is a chord of length 24 cm at a distance of 5 cm from the center, find the radius of the circle.

(a) 13 cm (b) 17 cm

(c) 25 cm (d) 15 cm

RRB JE - 28/06/2019 (Shift-III)

Ans. (a)



the perpendicular from the center of the circle on chord divides the chord into two equal parts.

$OA^2 = OC^2 + AC^2$ {where AC = 12 cm}

$r^2 = 5^2 + 12^2$

$r^2 = 25 + 144 = 169$

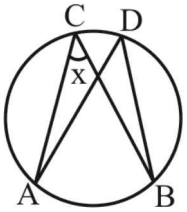
r = 13 cm

69. Read the given question and decide which of the following information is sufficient to answer the question.

What is the value of $\angle ACB$

Information:

(1)



(2) $\angle D = 60^0$

(a) Either 1 or 2 is sufficient.

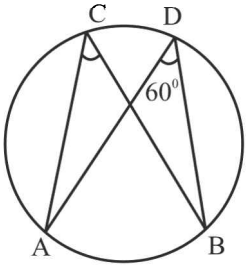
(b) Both 1 and 2 are sufficient.

(c) Only is 2 sufficient.

(d) Only 1 are sufficient.

RRB Group-D – 17/09/2018 (Shift-I)

Ans : (b) $\angle ADB = \angle ACB = 60°$



so both information will be required,

The value of the angle subtended by an arc (chord) on the remaining circumference of the circle is same.

$\angle C = \angle D$

Hence information-2, $\angle D = 60°$ and information -1, $\angle ACB = \angle ADB = 60°$

70. In a semicircle, A and C denote the diameter and B is a point on the semicircle. So the angle ABC will always be.

(a) Can not be determined

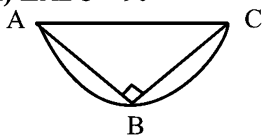
(b) Acute angle

(c) Obtuse angle

(d) Right angle

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (d) $\angle ABC = 90°$



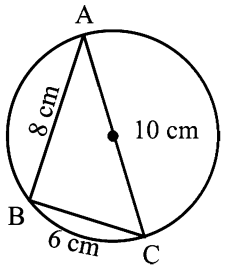
Theorem— The angle on the semicircle is right angled.

71. The area of the circumcircle of a right angled triangle with sides 6 cm, 8 cm and 10 cm will be–

(a) 16π cm^2 (b) 25π cm^2
(c) 24.5π cm^2 (d) 9π cm^2

RRB Group-D – 18/09/2018 (Shift-II)

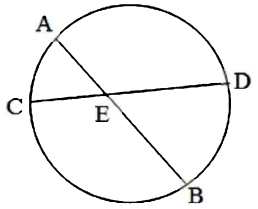
Ans. (b) :



$\therefore$ Circle's radius (r) = 5 cm

$\therefore$ area of circumcircle $= \pi r^2 = \pi \times (5)^2 = 25\pi$ cm^2

72.

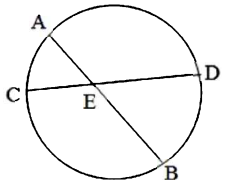


In above circle, given m $\overline{AE}$ = 4 cm, m $\overline{BE}$ = 15 cm and m $\overline{CE}$ = 2.5 cm. What will be the value of m $\overline{DE}$?

(a) 16.5 cm (b) 20 cm
(c) 24 cm (d) 30 cm

RRB Group-D – 24/10/2018 (Shift-I)

Ans : (c)



given–

m $\overline{AE}$ = 4 cm.

m $\overline{BE}$ = 15 cm

m $\overline{CE}$ = 2.5 cm

m $\overline{DE}$ = ?

$\boxed{AE \times BE = CE \times DE}$

$4 \times 15 = 2.5 \times DE$

$DE = \frac{4 \times 15}{2.5}$

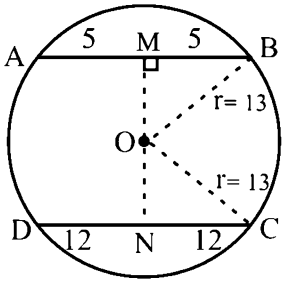
DE = 24 cm.

73. AB and CD are two parallel sides on opposite ends of the centre of the circle. If AB = 10 cm, CD = 24 cm and radius of the circle is 13 cm. Then what is the distance between the two sides.

(a) 16 cm (b) 17 cm
(c) 18 cm (d) 15 cm

RRB Group-D – 26/10/2018 (Shift-III)

Ans : (b)



from pythagoras theorem in ΔONC

$r^2 = ON^2 + NC^2$

$ON = \sqrt{r^2 - NC^2}$

r = 13 cm

$ON = \sqrt{13^2 - 12^2} = \sqrt{169 - 144}$

ON = 5 cm

like wise $MO = \sqrt{13^2 - 5^2} = 12$ cm

So distance between of both sine.

= MO + NO

= 12 + 5 = 17 cm

74. A 16 cm long arc is cut in a circle of 20 cm diamter. Find the distance of the arc from the center of the circle.

(a) 6 cm (b) 12.8 cm
(c) 12 cm (d) 18.9 cm

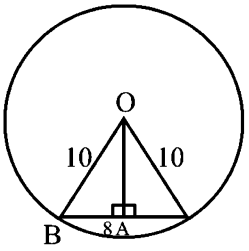
RRB Group-D – 16/11/2018 (Shift-I)

Ans : (a) diameter = 20 cm

arc cut by 16 cm

$\because$ The perpendicular drawn on the chord from the center of a circle bisects the chord.

$\therefore AB = \frac{16}{2} = 8$ cm



$OB^2 = OA^2 + AB^2$

$100 = OA^2 + 64$

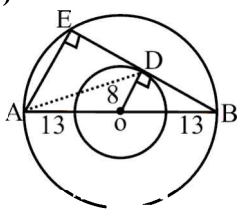
OA = 6 cm

75. The radius of two concentric circles are 13 cm and 8 cm. AB is the diameter of the larger circle, BD is the tangent line to the smallar circle which touches it on D. The length of AD is equal to:

(a) 17 cm (b) 18 cm
(c) 19 cm (d) 16 cm

RRB Group-D – 11/12/2018 (Shift-III)

Ans : (c)



Small or circle of tangent line is DB and tangent line of circle is making 90° angle from the center of circle on circumference.

So,

Given in, {OB = 13 cm, OD = 8 cm from pythagoras theorem}

In ΔODB,

$(BD)^2 = (OB)^2 - (OD)^2$

$BD = \sqrt{(13)^2 - (8)^2}$

$BD = \sqrt{169 - 64} = \sqrt{105}$

again in ΔAEB,

$EB = 2DB = 2\sqrt{105}$ cm

AB = 2OB = 26 cm

from the pythagoras theorem

$(AE)^2 = (AB)^2 - (EB)^2$

AB = 2OB = 26 cm

again in ΔAED

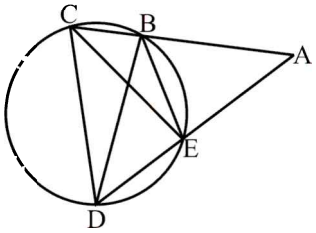
$(AD)^2 = (AE)^2 + (ED)^2$

$(AD)^2 = (16)^2 + \left(\sqrt{105}\right)^2$

$(AD) = \sqrt{256 + 105} = \sqrt{361}$

AD = 19 cm

76.

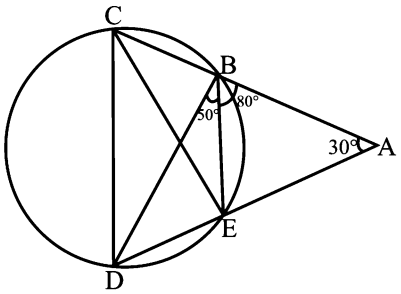


In the above figure, ∠BAE = 30°, ∠ABE = 80° and ∠DBE = 50°, what is the value of ∠BCE ?

(a) 25° (b) 10°
(c) 20° (d) 5°

RRB Group-D – 10/12/2018 (Shift-I)

Ans. (c) :



given

∠BAE = 30°

∠ABE = 80°

∠DBE = 50°

from ΔBAE

∠BEA + 80 + 30 = 180° [∵Sum of Δ of interior angles = 180°]

∴∠BEA = 180°–110° = 70°

∠DEB = 180° – 70° [angle of straight line = 180°]

= 110°

in cyclic quadrilateral CBED

∠DCB = 70° [∵Sum of opposite angle of cyclic quadrilateral = 180°]

∵∠DBE = 50° [angle formed on circumference from chord is equal]

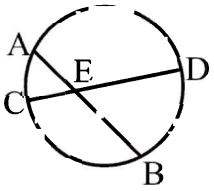
∴∠DCE = 50°

∵∠DCB = 70°

∴∠BCE = 70°–50° = 20°

So, ∠BCE = 20°

77.



In the above circle, $m\overline{AE} = 5cm, m\overline{BE} = 15cm$ and $m\overline{CE} = 25cm$, what will be the length of $m\overline{DE}$?

(a) 2.5 cm (b) 1 cm
(c) 3 cm (d) 2 cm

RRB Group-D – 23/10/2018 (Shift-II)

Ans. (c) : given

$m\overline{AE} = 5cm, m\overline{BE} = 15cm$

$m\overline{CE} = 25cm, m\overline{DE} = ?$

We know that,

If chord AB and CD is intersecting each other of inside the circle on point E, then

$$AE \times BE = CE \times DE$$

$$DE = \frac{AE \times BE}{CE}$$

$$= \frac{5 \times 15}{25} = 3cm$$

So, $m\overline{DE} = 3cm$.

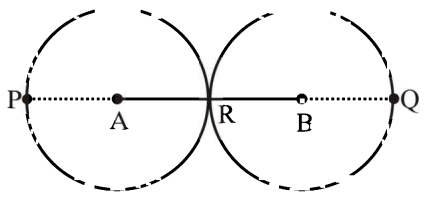
78. The radius of circle A and circle B is 4 units. If the point P is located on A and the point Q is located on B, and Both circle touch each other exactly at a point. Then what will be the maximum length of PQ.

(a) 0 (b) 4

(c) 8 (d) 16

RRB NTPC 30.03.2016 Shift : 2

Ans : (d)



Circle A and circle B touch on point R in the figure then,

AR = BR = 4 unit (radius of circle is)

PR then QR is diameter of circle A and circle B.

then PR = QR = 2AR

PR = QR = 2×4 = 8 unit

Or maximum length of PQ =2×PR = 2×8 = 16 unit

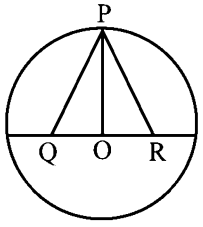
79. An equilateral triangle is constructed in such a way that the two end of the triangle are placed on the diameter of the circle and the third is placed on the circle. If the area of the circle is 48π then what will be the side of the triangle.

(a) 8 (b) 4

(c) $8/\sqrt{3}$ (d) $4\sqrt{3}$

RRB NTPC 10.04.2016 Shift : 3

Ans : (a) from the figure,



Let of circle radius = r

$\therefore$ area $= \pi r^2$

$= 48\pi = \pi r^2$

$r = \sqrt{48} = 4\sqrt{3}$

Radius = length of OP is perpendicular PQR equilateral triangle.

$$\Rightarrow 4\sqrt{3} = \frac{\sqrt{3}}{2}a$$

$$\therefore a = 4 \times 2 = 8$$

So length of triangle's sides = 8

80. If a circle is divided into 6 equal parts. What will be the measure of each angle?

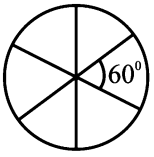
(a) 45 (b) 60

(c) 30 (d) 90

RRB NTPC 28.04.2016 Shift : 1

Ans : (b)

$\because$ angle formed by all 6 sides on the center of circle =360°



$\therefore$ According to the question

$$= \frac{360°}{6} = 60°$$

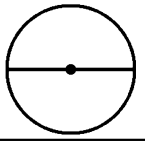
81. The largest chord of a circle is.

(a) Radius (b) Diameter

(c) Line segment (d) Sector

RRB NTPC 29.04.2016 Shift : 2

Ans : (b) Largest chord of circle is "diameter" which passes through the center of the circle.



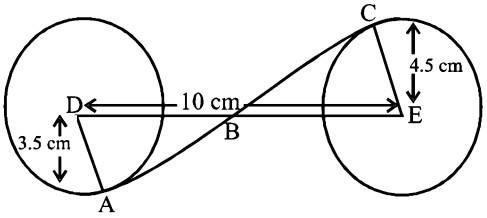
82. Find the length of a transverse common tangent of the two circles whose radii are 3.5 cm, 4.5 cm and the distance between their centres is 10 cm.

(a) 6 cm/6 cm (b) 8 cm/8 cm

(c) 6.4 cm/6.4 cm (d) 3.6cm/3.6 cm

RRB ALP & Tec. (17-08-18 Shift-II)

Ans : (a)



length of transverse tangent line

$$= \sqrt{(\text{distance between of centers})^2 - (\text{sum of both radius's})^2}$$

$$= \sqrt{10^2 - (3.5 + 4.5)^2} = \sqrt{100 - 64} = \sqrt{36} = 6 \text{ cm}$$

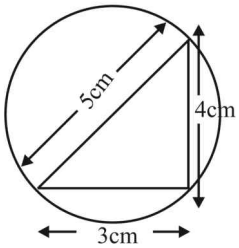
83. The two sides forming the right-angle in a triangle are 3 cm and 4 cm long respectively. What will be the area of the circumcircle of this tringle?

(a) $5\pi\ cm^2$ (b) $7\pi\ cm^2$

(c) $6.75\pi\ cm^2$ (d) $6.25\pi\ cm^2$

RRB ALP & Tec. (09-08-18 Shift-II)

Ans : (d)



The centre of the circumcircle formed on any right triangle is at the midpoint of the hypotenuse of the triangle, which is the diamter of the circle.

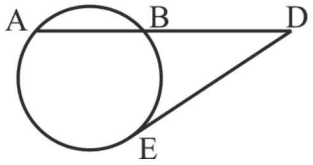
So radius of circle $= \frac{\sqrt{4^2+3^2}}{2} = \frac{\sqrt{25}}{2} = 5/2$ cm

area of circle $= \pi r^2$

$$= \pi \times \left(\frac{5}{2}\right)^2 = \pi \times \frac{25}{4} = 6.25\pi\ cm^2$$

Type - 9

84.



In the circle shown above, chord AB is extended so that it meets tangent line DE to D. If AB = 5 cm and DE = 6 cm, Find the length of BD.

(a) 6 cm (b) 5 cm

(c) 4 cm (d) $\sqrt{30}$ cm

RRB RPF Constable – 18/01/2019 (Shift-III)

Ans : (c) Since $DE^2 = DA \times BD$

$DE^2 = (AB+BD) \times BD$

$6^2 = (5+BD) \times BD$

$36 = 5BD + BD^2$

$\Rightarrow BD^2 + 5BD - 36 = 0$

$\Rightarrow BD^2 + 9BD - 4BD - 36 = 0$

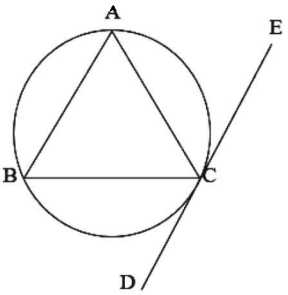
$\Rightarrow BD(BD+9) - 4(BD+9) = 0$

$(BD-4)(BD+9) = 0$

$BD - 4 = 0,\ BD \neq -9$

BD = 4 centimeter

85. Answer the question based on the figure given below.



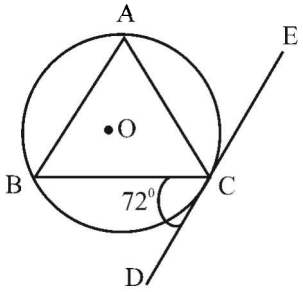
If AB is parallel to EC and ∠BCD = 72°, then what will be the value of ∠BAC?

(a) 75° (b) 72°

(c) 73° (d) 67°

RRB RPF SI – 05/01/2019 (Shift-II)

Ans : (b)



∠BCD = 72°

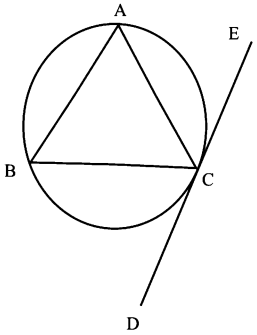
∠BAC = ?

If the angle between the line and the chord passing through one end of the chord of the circle is equal to the angle substended by the chord in the alternate segment, then this line is the tangent line of the circle.

Let the center of the circle be O and AC is the chord of this circle. A straight line DE going through the C point is drawn such that ∠BCD = ∠BAC where ∠BAC is loacted in alternating circle.

So, ∠BAC = 72°

86.



If ∠BCD = 82° then find the value of ∠BAC ?

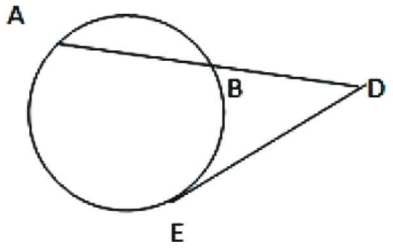
(a) 85° (b) 77°

(c) 83° (d) 82°

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (d) : The angle formed by the chord of a circle along the line of the circle is equal to the angle made by the chord on the alternate segment of the circle.
$\therefore \angle BAC = \angle BCD = 82°$

87. In the circle below, chord AB is extended to point D to join tangent DE. If AB = 9 cm and BD = 3 cm, find the length of DE.

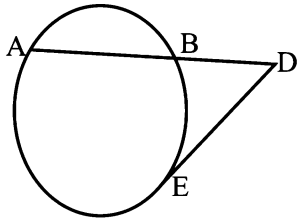


(a) 4 cm (b) $\sqrt{27}$ cm
(c) 6 cm (d) 5 cm

RRB Paramedical Exam – 21/07/2018 (Shift-I)

Ans : (c) from the theorem
$DE^2 = DB \times DA = 3 \times 12$
$DE^2 = 36$
$DE = 6$ cm

88. In the circle below, chord AB is extended to meet tangent DE at point D. If AB = 24 cm and DE = 9 cm, find the length of BD.

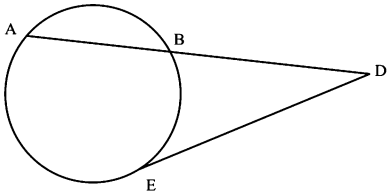


(a) 3 cm (b) $4\sqrt{6}$ cm
(c) 5 cm (d) 4 cm

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (a) : $AD \times BD = (DE)^2$
(Suppose $BD = x$)
$(24 + x) \times x = (9)^2$
$24x + x^2 = 81$
$x^2 + 24x - 81 = 0$
$x^2 + 27x - 3x - 81 = 0$
$x(x + 27) - 3(x + 27) = 0$
$x + 27 = 0, x = -27$ (Negative)
$x - 3 = 0$
$x = 3$
length of BD = 3 cm

89. In the circle below, chord AB is extended to join point D with tangent DE. If AB = 6 cm and BD = 2 cm, find the length of DE.



(a) $\sqrt{12}$cm (b) 4cm
(c) 4.5cm (d) 5 cm

RRB Group-D – 05/10/2018 (Shift-I)

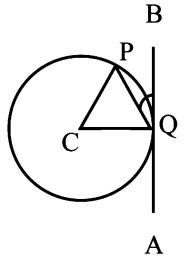
Ans. (b) : from given figure we know that
$\therefore DE^2 = AD \times BD$
$DE^2 = (AB + BD) \times BD$
$= (6 + 2) \times 2$
$= 8 \times 2$
$DE^2 = 16$
$DE = 4$ cm

90. P is a point on a circle whose center is C. A straight line whose length is equal to the radius of the circle drawn from P to another point Q. Find the value of the acute angle in radian Q formed by the tangent of the circle drawn on P.

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{3}$

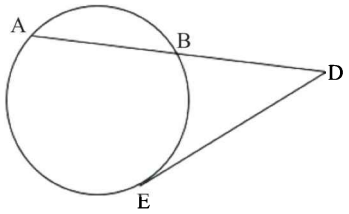
RRB Group-D – 10/10/2018 (Shift-I)

Ans : (a)



from the figure– CP = CQ
given– radius r = PQ
$\therefore$ CP = CQ = PQ (equilateral triangle)
$\because$ Value of each angle of equilateral trianle = 60°
So value of acute angle on $\angle PQB$
$= 90° - 60° = 30° = \frac{\pi}{6}$ radian

91.



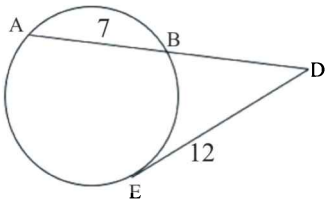
In the given circle, the line AB is extended to meet with the tangent DE at the D point. If AB = 7 cm and DE = 12 cm, then the length of BD will be?

(a) 7 cm (b) 9 cm

(c) $2\sqrt{21}$ cm (d) 8 cm

RRB Group-D – 24/10/2018 (Shift-I)

Ans : (b)



$DE^2 = AD \times BD$

If BD = x then

$12^2 = (7 + x) \times x$

$x^2 + 7x - 144 = 0$

$x^2 + 16x - 9x - 144 = 0$

$x(x+16) - 9(x+16) = 0$ or $x + 16 = 0$, $x - 9 = 0$ hence

$x = 9$, $x = -16$ (invalid)

So, BD = x = 9 cm

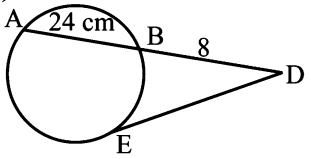
92. In a circle, chord AB is extended to touch tangent DE to D. If AB = 24 cm and BD = 8 cm, then the length of DE is known.

(a) 12 cm (b) 18 cm

(c) 16 cm (d) 24 cm

RRB Group 'D' 07/12/2018 (Shift-I)

Ans : (c)



$DE^2 = AD \times BD$

$DE^2 \ (24 + 8) \times 8$

$DE^2 = 32 \times 8$

$DE = \sqrt{16 \times 16}$

DE = 16 cm.

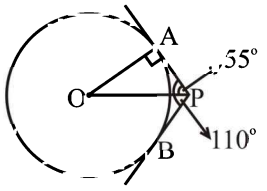
93. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 110^0 then the angle POA is :

(a) 50° (b) 70°

(c) 35° (d) 45°

RRB ALP & Tec. (20-08-18 Shift-III)

Ans : (c)



ΔPOA in $\angle OAP = 90^o$ (put perpendicular on tangent line make right angle.)

$\angle OPA = 55^o \quad \left(\frac{110^o}{2} = 55^o\right)$

So $\angle OAP + \angle OPA + \angle POA = 180^o$

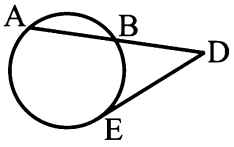
$90^o + 55^o + \angle POA = 180^o$

$\angle POA = 180^o - (90^o + 55^o)$

$= 180^o - 145^o$

$\angle POA = 35^o$

94.



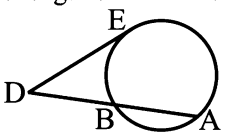
In the circle above chord $\overline{AB}$ is extended to meet the tangent $\overline{DE}$ at D. If $\overline{AB}$ = 12 cm and $\overline{DE}$ = 8 cm. Find the length of $\overline{BD}$.

(a) 6 cm (b) $\sqrt[4]{6}$ cm

(c) 5 cm (d) 4 cm

RRB ALP & Tec. (09-08-18 Shift-I)

Ans : (d) Suppose length of BD = x cm



formula-$(DE)^2 = DB \times DA$

$(8)^2 = x \times (x + 12)$

$\Rightarrow$ $64 = x^2 + 12x$

$\Rightarrow$ $x^2 + 12x - 64 = 0$

$\Rightarrow$ $x^2 + (16 - 4)x - 64 = 0$

$\Rightarrow$ $x^2 + 16x - 4x - 64 = 0$

$\Rightarrow$ $x(x + 16) - 4(x + 16) = 0$

$\Rightarrow$ $(x + 16)(x - 4) = 0$

$\Rightarrow$ $x = -16$ (invalid)

$\Rightarrow$ $x = 4$ cm.

So, DB = 4 cm.

95. To draw a pair of tangents to a circle which are inclined to each other at an angle of 75°, it is required to draw tangents at the end points of those two radius of the circle, the angle between then is.

(a) 65° (b) 75°
(c) 95° (d) 105°

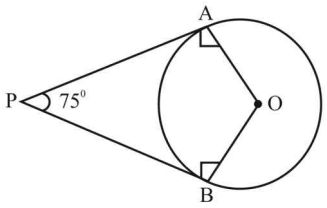
RRB ALP & Tec. (09-08-18 Shift-II)

Ans : (d) from the figure,

$\angle PAO = \angle PBO = 90°$

$\angle APB = 75°$

$\because$ PAOB is a quadrilaterals



$\because$ Sum of four angles of quadrilaterals is 360°

$\because$ PAOB is a quadrilaterals–

$\therefore \angle APB + \angle PBO + \angle PAO + \angle AOB = 360°$

$75° + 90° + 90° + \angle AOB = 360°$

$\boxed{\angle AOB = 360° - 255° = 105°}$

Type - 10

96. In the second quartile, what is the value of sinθ between?

(a) –1 to 1 (b) 1 to –1
(c) 1 to 0 (d) 0 to 1

RRB JE - 30/05/2019 (Shift-I)

Ans : (c) Second quarter is between 90° from 180°

$\sin 90° = 1, \quad \sin 180° = 0$

Similar- $0 < \sin\theta < 1$

Value of sin θ is between from 1 to 0.

97. The ratio of interior and exterior angles of a regular polygon is 4:1. What is the number of sides in the polygon?

(a) 12 (b) 6
(c) 8 (d) 10

RRB RPF Constable – 24/01/2019 (Shift-III)

Ans : (d) Suppose interior angle and exterior angle is 4x and x.

Interior angle = 180 - exterior angle

$4x = 180 - x$

$5x = 180$

$x = 36$

$$\text{number of sides} = \frac{360}{\text{exterior anlge}} = \frac{360}{36} = 10$$

98. What will be the sum of the interior angles in degrees of a polygon with 7 sides?

(a) 180° (b) 360°
(c) 540° (d) 900°

RRB RPF Constable – 20/01/2019 (Shift-III)

Ans : (d) Sum of interior angles of polygon

$= (2n-4) \times 90° \quad = (2 \times 7-4) \times 90°$

$= 10 \times 90° \quad = 900°$

99. Each interior angle in a regular polygon is 108°, then what is the numberof sides?

(a) 6 (b) 15 (c) 5 (d) 7

RRB RPF Constable – 25/01/2019 (Shift-I)

Ans : (c) Number of sides are n then

each intenior angle of regular polygon $= \frac{(n-2)\times 180}{n}$

$$108 = \frac{(n-2)\times 180}{n}$$

$108n = 180n - 360$

$180n - 108n = 360$

$72n = 360$

$$n = \frac{360}{72}$$

Number of sides $n = 5$

100. What will be the total sides of a regular polygon whose exterior angle is 10°?

(a) 36 (b) 63 (c) 46 (d) 38

RRB RPF SI – 06/01/2019 (Shift-II)

Ans : (a) each exterior angle of polygon

$$= \frac{360^0}{n\,(\text{number of sides})}$$

$$n = \frac{360}{10} = 36$$

101. The sum of all interior angles of a common polygon is 1440°. How many diagonals does the polygon have?

(a) 27 (b) 44
(c) 35 (d) 20

RRB RPF Constable – 22/01/2019 (Shift-II)

Ans : (c) n sum of polygon's interior angles of n sides

$(\Sigma\theta) = n \times 180 - 360$

$1440 = n \times 180 - 360$

$n \times 180 = 1440 + 360$

$n \times 180 = 1800$

$n = 10$

If n sides is in any polygon then number of their diagonals $D = \frac{n}{2}(n-3)$

So number of these diagonals.

$$D = \frac{10}{2}(10-3)$$

$D = 5 \times 7$

$D = 35$

102. The interior angle of a regular polygon is 150°. This polygon is a ___.

(a) Octagon (b) Decagon
(c) Dodecagon (d) Heptagon

RRB JE - 22/05/2019 (Shift-I)

Ans : (c)
each interior angle of a regular polygon with n sides.

$$= \frac{(n-2)\times 180}{n}$$

$$150 = \frac{(n-2)\times 180}{n}$$

150n = 180n – 360

30n = 360

$$n = \frac{360}{30} = 12$$

Therefore this polygon will be a dodecagon.

103. Select the incorrect statement

1) All sides of a regular polygon are equal.
2) Its all interior angles are equal.
3) The sum of its exterior angles is 360°
4) The sum of its interior angles are (n–2) × 360°

(a) Statement 1 (b) Statement 4
(c) Statement 2 (d) Statement 3

RRB JE - 29/05/2019 (Shift-II)

Ans : (b) Sum of regular polygon's interior angles

= (n – 2) × 180

Hence all other statement is true when statement (4) is false.

104. Each interior angle of a regular polygon is 36° greater than its exterior angle. Find out the number of sides of polygon–

(a) 4 (b) 10
(c) 8 (d) 5

RRB JE - 02/06/2019 (Shift-I)

Ans : (d) Suppose number of sides of polygon = n
As per the question,

$$\frac{(2n-4)\times 90}{n} - \frac{360°}{n} = 36°$$

180n - 360° – 360° = 36n

144n = 720°

$$n = \frac{720}{144}$$

Hence number of sides = 5

105. The sum of the angles of regular polygon is 2160°. The number of sides of polygon is–

(a) 12 (b) 16
(c) 18 (d) 14

RRB JE - 27/06/2019 (Shift-I)

Ans : (d) Sum of interior angles of regular Polygon.

= 2160°

(2n – 4) 90° = 2160°

(2n – 4) = 24

2n = 28

n = 14

106. What is the total number of sides of regular polygon with each interior angle equal to 168°?

(a) 20 (b) 30
(c) 15 (d) 31

RRB RPF SI – 12/01/2019 (Shift-II)

Ans : (b) Each interior angle of regular polygon

$$= \frac{(n-2)\pi}{n}$$

$$\frac{(n-2)180^0}{n} = 168^0$$

180°n–360°=168°n

12°n = 360°

n = 30

So, number of sidess (n) = 30

107. A polygon with 7 sides is called_____

(a) Nonagon (b) Hexagon
(c) Heptagon (d) Octagon

RRB NTPC 28.03.2016 Shift : 1

Ans : (c) A polygon with 7 sides is called heptagon.

108. The sum of the interior angles of a polygon is equal-

(a) (n – 2) 360 (b) (n – 3) 180
(c) (n – 2) 180 (d) (n + 2) 180

RRB NTPC 07.04.2016 Shift : 1

Ans : (c)
Sum of interior angles of polygon is (n –2) 180

109. In a polygon, each exterior angle is 120°, then the number of sides is-

(a) 6 (b) 4
(c) 3 (d) 5

RRB NTPC 07.04.2016 Shift : 3

Ans :(c)

$$\text{Number of sides of polygon} = \frac{360°}{\text{each exterior angle}}$$

$$= \frac{360}{120} = 3$$

110. In a regular polygon, each exterior angle is 60° then the number of sides:

(a) 7 (b) 5
(c) 6 (d) 8

RRB NTPC 26.04.2016 Shift : 2

Ans : (c) each exterior angle of polygon $= \frac{360^0}{n}$

$\therefore \quad n = \frac{360^0}{60^0} = 6$

111. The number of diagonals in a polygons with 27-sides is-

(a) 320 (b) 324
(c) 322 (d) 325

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (b) : The number of diagonals in a polygon with n-sides

$= \frac{n}{2} \times (n-3)$

$\because n = 27$

$\therefore$ number of diagonal $= \frac{27}{2} \times (27-3)$

$= \frac{27 \times 24}{2} = 27 \times 12 = 324$

112. The sum of all interior angles of a polygon is 1260°. Find the number of sides of the polygon.

(a) 9 (b) 8 (c) 10 (d) 11

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (a) : Suppose numbr of sides of palygon = n

According to the question-

$(n-2)\pi = 1260, \quad (n-2)180 = 1260$

$n-2 = 7, \quad n = 7+2, \quad \boxed{n=9}$

113. The sum of the interior angles of a regular octagon is___

(a) 1080 (b) 720
(c) 540 (d) 900

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (a) Sum of interior angle of an regular polygon with n sides = (2n – 4) × 90°

∴ Sum of interior angle of an octagon = (2 × 8 – 4) × 90 = 1080

114. The measure of each interior angle of a pentagon is _____.

(a) 180^0 (b) 108^0
(c) 115^0 (d) 120^0

RRB Group-D – 05/10/2018 (Shift-II)

Ans. : (b) Exterior angle of any polygon,

$= \frac{360^0}{\text{number of sides}}$

exterior angle $= \frac{360^0}{5} = 72^0$

interior angle = 180° – 72° = 108°

115. What is the diagonal's number in a 19-sides?

(a) 304 (b) 114
(c) 152 (d) 76

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (c) Number of diagonals in polygon with n sides

$= \frac{n(n-1)}{2} - n = \frac{19 \times (19-1)}{2} - 19$

$= \frac{19 \times 18}{2} - 19 = 19 \times 9 - 19$

= 19 (9 – 1) = 19 × 8 = 152

116. What will be the ratio of the measurement of the interior angles of a regular octagon to the interior angles of a regular dodecagon?

(a) 4 : 5 (b) 8 : 12
(c) 12 : 8 (d) 9 : 10

RRB Group-D – 12/10/2018 (Shift-III)

Ans : (d) Sum of interior angles of polygon with n sides = $(2n-4) \times 90^0$

each interior angle of octagon $= \frac{(2 \times 8 - 4) \times 90^0}{8}$

$= \frac{12 \times 90}{8} = \frac{1080}{8}$

each interior angle of dodecagon $= \frac{(2 \times 12 - 4) \times 90^0}{12}$

$= \frac{1800}{12}$

ratio $= \frac{1080}{8} : \frac{1800}{12} = 9:10$

117. The number of diagonals in a polygon of 26-sides will be–

(a) 325 (b) 300
(c) 299 (d) 650

RRB Group-D – 25/09/2018 (Shift-I)

Ans : (c) Diagonals number of polygon $= \frac{n}{2} \times (n-3)$

number of diagonals in polygon with 26 sides

$= \frac{26}{2}(26-3) = 13 \times 23 = 299$

118. The number of diagonals a polygon with in 19-sides will be –

(a) 161 (b) 133
(c) 171 (d) 152

RRB Group-D – 25/09/2018 (Shift-III)

Ans. (d) : number of diagonals $= \frac{n}{2}(n-3)$

$= \frac{19}{2}(19-3) = \frac{19}{2} \times 16$

$= 19 \times 8 = 152$

119. The measure of each interior angle of a normal octagon is–

(a) 115^0 (b) 125^0
(c) 120^0 (d) 135^0

RRB Group-D – 04/10/2018 (Shift-II)

Ans : (d) Measure of interior angles of polygon

$$=\frac{(n-2)180}{n}$$

$$=\frac{(8-2)\times 180}{8} = 135^0$$

120. The value of each interior angle of a regular heptagon is _____

(a) 128.57^0 (b) 126.37^0
(c) 148.24^0 (d) 137.56^0

RRB Group-D – 05/10/2018 (Shift-I)

Ans. (a) : ∵ each interior angle of regular polygon with n sides

$$=\frac{(2n-4)\times 90}{n}$$

∴ each interior angle of regular heptagon

$$=\frac{(2\times 7-4)\times 90}{7}$$

$$=\frac{900}{7}=128.57^0$$

121. What is the value of each interior angle of a regular pentagon?

(a) 148.24° (b) 137.56°
(c) 128.57° (d) 108°

RRB Group-D – 22/10/2018 (Shift-III)

Ans : (d) Value of interior angle of polygon with n sides

$$= \frac{(n-2)}{n}180$$

each interior angle of regular pentagon

$$= \frac{(5-2)}{5}180$$

$$= \frac{3\times 180}{5} = \frac{540}{5} = 108°$$

122. The number of diagonals in a polygon with 28 sides is–

(a) 350 (b) 280
(c) 304 (d) 175

RRB Group-D – 11/12/2018 (Shift-I)

Ans. (a) : number of sides (n) = 28

Number of diagonals $=\frac{n(n-3)}{2}$

$$=\frac{28(28-3)}{2} = \frac{28\times 25}{2} = 350$$

123. What is the number of diagonals in a polygon with 17-sides:–

(a) 118 (b) 120
(c) 121 (d) 119

RRB Group-D – 31/10/2018 (Shift-III)

Ans : (d) Number of diagonals in Polygon $= \frac{n}{2}(n-3)$

$$=\frac{17}{2}(17-3) = \frac{17}{2}\times 14 = 119$$

124. A polygon has 9 sides. What will be its interior angle?

(a) 140^0 (b) 100^0
(c) 120^0 (d) 40^0

RRB NTPC 30.03.2016 Shift : 1

Ans : (a) each interior angle of regular polygon—

$$=\frac{(2n-4)\times 90^0}{n} = \frac{(2\times 9-4)\times 90^0}{9} = 140^0$$

125. The number of sides of a regular polygon whose internal angle is 150° will be–

(a) 15 (b) 13
(c) 12 (d) 14

RRB NTPC 17.01.2017 Shift-3

Ans : (c) Let number of sides = n

each interior angle of regular polygon $=\frac{(2n-4)90^0}{n}$

$$150=\frac{180n-360}{n}$$

150 n = 180n – 360

$30n = 360^0 \Rightarrow n = 12$

126. If each of the exterior angle of a regular polygon is 9°, then the number of sides of regular polygon will be–

(a) 30 (b) 40
(c) 45 (d) 36

RRB NTPC 17.01.2017 Shift-1

Ans : (b) each exterior angle of regular polygon

$$=\frac{360}{n}$$

So, $=\frac{360^0}{n}=9^0$

$n=\frac{360^0}{9^0}=40$ sides

127. Each exterior angle in a polygon is 72°. The number of sides is–

(a) 7 (b) 6
(c) 5 (d) 8

RRB NTPC 11.04.2016 Shift : 1

Ans : (c) Number of sides in a polygon

$$=\frac{360^0}{\text{each exterior angle}} = \frac{360^0}{72^0} = 5$$

128. In a regular polygon, each exterior angle is 36°. Find the number of its sides.

(a) 11 (b) 9
(c) 10 (d) 8

RRB NTPC 07.04.2016 Shift : 2

Ans : (c) In polygon

each exterior angle $= \frac{360^0}{\text{number of sides}}$

$\Rightarrow 36^0 = \frac{360^0}{\text{number of sides}}$

$\therefore$ Number of sides = 10

129. In a polygon, each exterior angle is 45°, then the number of sides is:–

(a) 7 (b) 6
(c) 8 (d) 9

RRB NTPC 22.04.2016 Shift : 2

Ans : (c) Number of sides of polygon

$= \frac{360°}{\text{each exterior angle}} = \frac{360}{45} = 8$

130. In a polygon, each exterior angle is 40°, then the number of sides is:

(a) 8 (b) 45
(c) 9 (d) 10

RRB NTPC 26.04.2016 Shift : 3

Ans : (c) Number of sides $= \frac{360°}{\text{each exterior angle}}$

$= \frac{360°}{40} = 9$

131. What is the number of sides of a regular congruent triangle?

(a) 6 (b) 4
(c) 3 (d) 2

RRB NTPC 30.04.2016 Shift : 3

Ans : (c) Sides number of any triangle is 3 and sum of three interior angle is 180°.

Type - 11

132. If two complementary angles are in the ratio of 11:7, then find the smaller angle.

(a) 35° (b) 55°
(c) 45° (d) 25°

RRB RPF SI – 11/01/2019 (Shift-II)

Ans : (a) Sum of two complementary angle is 90°

Small angle $= \frac{7}{11+7} \times 90 = \frac{7}{18} \times 90 = 35^o$

133. The supplement of an angle is 15° more than three times its complementary. What will be the value of the angle?

(a) 57.5° (b) 72.5°
(c) 52.5° (d) 65°

RRB Group-D – 16/10/2018 (Shift-III)

Ans : (c) Suppose angle = x°

According to the question

$180° - x° = 3(90° - x°) + 15°$

$2x° = 270° + 15° - 180°$

$2x° = 105°$

$\boxed{x = 52.5°}$

134. How many degrees does an angle that is 1/5 of its supplementary angle?

(a) 45° (b) 30°
(c) 60° (d) 75°

RRB NTPC 04.04.2016 Shift : 2

Ans : (b) Suppose an angle = x°

its supplementary angle = 5x°

sum of two supplementary angle = 180°

$x + 5x = 180°$

$6x = 180 \Rightarrow x = 30°$

required angle = 30°

135. If two supplementary angles are in the ratio 4:5, Find the larger angle.

(a) 80° (b) 50°
(c) 60° (d) 100°

RRB NTPC 16.04.2016 Shift : 1

Ans : (d) Suppose angle is 4x° and 5x° respectively

We know that sum of two suuplemetary angle is 180°.

$\therefore 4x° + 5x° = 180°$

$9x° = 180°$

$x° = 20°$

$\therefore$ large angle $(5x°) = 5 \times 20° = 100°$

136. If the ratio of two supplementary angles are 13:5, find the difference between the two.

(a) 60° (b) 70°
(c) 80° (d) 120°

RRB NTPC 18.04.2016 Shift : 2

Ans : (c) Let first angle = 13x°

Second angle = 5x°

then $13x° + 5x° = 180°$

(sum of supplementary angle)

$18x° = 180°$, $x° = 10°$

first angle = 13 × 10° =130°

second angle = 5 × 10° = 50°

required difference = 130° – 50° = 80°

137. If (6y+70)° and (3y+47)° are supplementary angles, find the value of y.

(a) 12 (b) 15

(c) 7 (d) 10

RRB NTPC 12.04.2016 Shift : 2

Ans : (c) $(6y + 70)° + (3y + 47)° = 180°$

$6y° + 70° + 3y° + 47° = 180°$

$9y° + 117° = 180°$

$9y° = 180° – 117°$

$9y° = 63°$, $y° = 7°$

138. If (7x + 5)° and (x + 5)° are complementary angles, then find value of x.

(a) 10°

(b) 20°

(c) 30°

(d) 40°

RRB NTPC 12.04.2016 Shift : 3

Ans : (a) We know that sum of complementary angles is equal to 90°

∴ According to the question,

$(7x + 5)° + (x+5)° = 90°$

$\Rightarrow 7x° + 5° + x° + 5° = 90°$

$\Rightarrow 8x° = 90° – 10°$

$8x° = 80°$

$x° = \frac{80°}{8°} = 10°$

139. The ratio of two complementary angles are 4:5, find the ratio of the square of the first angle to the square of the second angle.

(a) 16 : 25

(b) 64 : 125

(c) 100 : 125

(d) 25 : 16

RRB NTPC 28.04.2016 Shift : 2

Ans : (a) Suppose both complementary angle are 4x° and 5x° respectively

$\therefore 4x° + 5x° = 90°$

$9x° = 90°$

$x° = 10°$

∴ first complementary angle = 40°

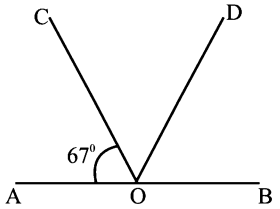
second complementary angle = 50°

∴ required ratio = $(40)^2 : (50)^2$

= 1600 : 2500 = 16 : 25

Type - 8

140. In the given figure, AOB is a straight line, ∠AOC = 67⁰ and the bisector of ∠BOC is OD. What is the value of ∠BOD in degrees?



(a) 56° (b) 55.5°

(c) 55° (d) 56.5°

RRB Group-D – 20/09/2018 (Shift-III)

Ans : (d) ∠AOC = 67°

∴∠BOC = 180 – 67

∠BOC = 113°

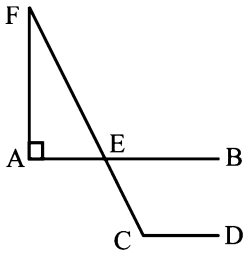
Bisector of ∠BOC is OD

So, $\angle BOD = \frac{\angle BOC}{2}$

$\angle BOD = \frac{113}{2}$

∠BOD = 56.5°

141. In the figure, AB || CD and ∠AFE = 30° then ∠FCD is:

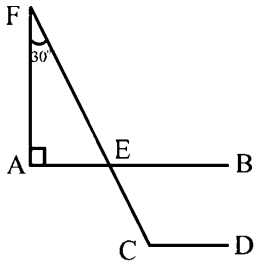


(a) 60° (b) 120°

(c) 90° (d) 45°

RRB Group-D – 05/11/2018 (Shift-III)

Ans. (b) :



Given,

AB || CD

∠AFE = 30°

∠FCD = ?

ΔAEF से

$\angle AFE = 30°$ (given)

$\angle FAE = 90°$

$\because$ sum of all three angles of Δ is 180°

$\therefore 90° + 30° + \angle AEF = 180°$

$\therefore \angle AEF = 180° - 120° = 60°$

$\angle BEC = 60°$ (vertically opposite angle)

$\angle FEB + \angle BEC = 180°$ (alternate angle)

$\angle FEB + 60° = 180°$

$\angle FEB = 120°$

$\because \angle FEB = \angle FCD$ (corresponding angle)

$\therefore \angle FCD = 120°$

142. There are two vertical pillars of heights 8 m and 12 m. One rope is drawn from the top of both the pillars to the bottom of the other. At what height from the ground do the ropes cut each other?

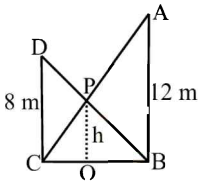
(a) 24/5 m (b) 22/7 m

(c) 29/6 m (d) 31/8 m

RRB Group-D – 12/10/2018 (Shift-I)

Ans. (a) : given-

heights of pillars = 8 m, 12 m

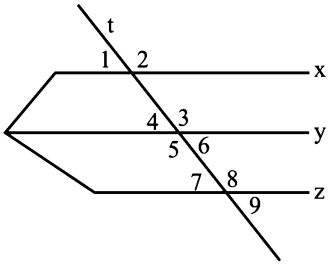


The rope cuts each other at a height of h meter from the ground.

from, $OP = \frac{a \times b}{a + b}$

$h = \frac{12 \times 8}{12 + 8} = \frac{12 \times 8}{20} = \frac{24}{5}$ m

143.



x, y and z are parallel lines and t is a transversal intersecting them all. Which of the following lists of angles is the same?

(a) 2, 3, 5 (b) 2, 6, 8

(c) 1, 4, 9 (d) 4, 5, 7

RRB NTPC 31.03.2016 Shift : 1

Ans : (c) $\angle 1 = \angle 4$ (Corresponding angle)

$\angle 4 = \angle 6$ (Vertical opposite angle)

$\angle 6 = \angle 9$ (Corresponding angle)

So angles 1, 4, 9 are equal/same

144. Consider the following statements, and choose the correct option:–

When two straight lines intersect each other, then

1. Adjacent angles are complementary.

2. Opposite angles are complementary.

(a) Both 1 and 2 are incorrect.

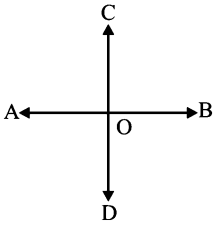
(b) Both 1 and 2 are correct.

(c) 1 is incorrect and 2 is correct.

(d) 1 is correct and 2 is incorrect.

RRB NTPC 16.04.2016 Shift : 1

Ans : (d)



When two straight lines intersect each other then adjacent angle be complementary angle.

145. Horizontal lines are on the same line.

(a) Equal in length

(b) Parallel to each other

(c) Intersecting each other

(d) From a triangle

RRB NTPC 29.04.2016 Shift : 1

Ans : (b) horizontal line on one line is parallel of each other.